

The Sentinel

VOL. 1, NO. 042

2026-03-30

COVER STORY

Starvation Mode: How Caloric Restriction Affects Your Body

Yep, starvation mode is a thing. But it doesn't have to halt weight loss.

Greatist

Cutting back on calories but not losing weight? You might wonder if you've gotten stuck in the twilight zone of so-called "starvation mode."

Everyone from your personal trainer to your grandma may have warned you about this metabolic frenemy that keeps your body holding onto weight. But is starvation mode really a thing — or just an urban legend?

The concept of starvation mode is that when you restrict calories, your body has a "hold it right there" response. To prevent you from losing weight rapidly, your metabolism slows, conserving energy by reducing the calories it burns.

Research shows starvation mode — formally called adaptive thermogenesis or metabolic adaptation — is a legit phenomenon. It's well documented that sometimes, the more weight you lose (and the more rapidly you do so), the more your body resists the process.

Evolutionarily speaking, this may be a holdover from when humans faced food scarcity and needed a mechanism for self-preservation. But in today's world, starvation mode can be a frustrating roadblock to weight loss.

What does this metabolic hiccup look like? These symptoms may indicate that starvation mode is becoming a problem for your health:

fatigue

hair loss

irritability

constipation

extreme hunger

feeling unusually cold

decreased muscle mass

a weight loss plateau with no change in diet or exercise habits

Starvation mode isn't the same thing as actual starving. In today's modern food environment, true starvation is rare. If you're eating regularly, you won't experience life-threatening starvation.

However, your body might exhibit some of the same early signs as real starvation. These include the symptoms listed above.

Most commonly, though, starvation mode involves stalled weight loss. You may have hit the wall of adaptive thermogenesis when restricting calories and increasing physical activity without losing weight.

How long until your body goes into starvation mode?

Getting into starvation mode can take just a few days, especially if those days involve severe calorie restriction.

So what counts as severe calorie restriction? Many experts advise not going below 1,200 calories per day for women and 1,500 calories per day for men unless supervised by a doctor. Anything below these numbers could land you in starvation mode faster.

How do I get my body out of starvation mode?

Fortunately, getting your body out of starvation mode isn't complicated.

First, there's the obvious solution: Eat a few more calories. But we don't mean just any calories. By adding a few more nutrient-dense foods (especially those with fiber and protein), you may find weight loss becomes easier, not harder. Both of these nutrients have links to weight loss — so they might help you get out of starvation mode without gaining weight.

If you're feeling extremely depleted, you might take a temporary break from a weight loss diet to replenish your nutrient stores.

Choosing the proper exercise could also bump you out of starvation mode. Resistance exercises like weightlifting, yoga, push-ups, squats, and others build muscle, which increases your metabolic rate.

According to a 2014 study , people who participated in a 9-month resistance exercise program increased their resting metabolic rate — aka how many calories they burned at rest — by an average of 5%.

When to see a doctor

Weight loss is a balancing act. It's normal to experience physical and emotional ups and downs.

That said, it's time to see a doctor if you're noticing symptoms like extreme fatigue or lethargy, constipation that lasts more than a few days, sensitivity to cold, or unusual mood changes. Your healthcare team can help you hit the sweet spot of weight loss without lingering in starvation mode.

Despite its name, being in starvation mode doesn't mean you'll starve. Instead, it's a state in which your body has adapted to your weight loss efforts, giving you pushback on how many calories it burns. With strategies like tweaking your calorie intake and adding strength training, you can get out of this mode and back on the path to sustained weight loss.

You may have hit the wall of adaptive thermogenesis when restricting calories and increasing physical activity without losing weight.

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FEATURES

Stopping Ozempic, Wegovy May Reverse Cardiovascular Benefits

Healthline

Stopping GLP-1s can quickly reverse the cardiovascular benefits gained while taking them. Image Credit: the_burtons/Getty Images

A recent study found that stopping GLP-1s, such as Ozempic or Wegovy, can reverse the cardiovascular benefits they provide.

The findings show that stopping the medications for as little as 6 months raises the risk of heart attack and stroke.

GLP-1s have been proven to offer not only benefits for type 2 diabetes and weight loss, but also cardiovascular health.

GLP-1 drugs like Ozempic and Wegovy have become popular medications for treating type 2 diabetes and obesity. This class of medications may also offer significant cardiovascular benefits.

A recent study published in *BMJ Medicine* found that when people stop using GLP-1s, they not only tend to regain weight, but they also may experience an increased risk of heart attack, stroke, and even death.

Around 1 in 8 adults in the United States is currently taking a GLP-1 medication.

“There is enormous exuberance about starting GLP-1 drugs, but not nearly enough attention to what happens when people stop,” senior study author Ziyad Al-Aly, MD, a Washington University School of Medicine clinical epidemiologist and chief of the Research and Development Service at the VA Saint Louis Health Care System, said in a press release.

Stopping GLP-1s raises cardiovascular risk

The researchers noted that many people who use these medications quit them after a short time, typically due to cost, side effects, or shortages.

They wanted to understand the consequences of discontinuing GLP-1 use, particularly on cardiovascular health.

The study analyzed 333,687 veterans. It compared 132,551 individuals who were prescribed a GLP-1 medication to help manage type 2 diabetes with 201,136 who were prescribed sulfonylureas, another type of medication for diabetes. The researchers followed the participants’ outcomes for 3 years.

Sulfonylureas include the medications:

glipizide (Glucotrol)

glimepiride (Amaryl)

glyburide (Diabeta and others)

The researchers checked participants’ GLP-1 treatment status every 6 months.

Over the course of the study, 26% of participants stopped taking the medication, and 23% had an interruption of 6 months or more, followed by resuming treatment.

The research team found a positive relationship between continuous use of GLP-1s and fewer cardiovascular events.

“GLP-1 drugs likely help cardiovascular health through several pathways at once, not just by lowering weight,” said Robert Glatter, MD, attending physician in the Department of Emergency Medicine at Lenox Hill Hospital in New York City, and assistant professor of Emergency Medicine at Zucker School of Medicine at Hofstra/ Northwell, who was not involved in the study.

“They improve blood sugar control, modestly lower blood pressure, may improve lipid and vascular function, and seem to reduce inflammation and atherosclerotic plaque growth and progression,” Glatter told Healthline.

“Some evidence also points to direct protective effects on the heart and blood vessels independent of weight loss. In practical terms, they appear to reduce the underlying process of inflammation that drives heart attacks, strokes, and heart failure over time,” he explained.

At the end of the study, compared with those who took sulfonylureas, participants who continuously used GLP-1s over the 3-year period had the most pronounced risk reduction. This group saw 18% fewer major cardiovascular events.

Participants who had taken GLP-1s for 2 or 2.5 years before discontinuing use for the remainder of the study also saw a significant reduction in risk of 7% and 15%, respectively.

Those who took GLP-1s for 18 months or less before discontinuing did not experience a significant reduction in risk.

The study showed that an interruption of GLP-1 use of just 6 months before resuming treatment was enough to significantly decrease the cardiovascular benefit. It led to a 4% to 8% increase in risk compared with those with continuous use.

Discontinued use of 1 to 2 years without resuming resulted in a 14% to 22% increased risk of a cardiovascular event, compared with continuous use.

This shows that cardiovascular benefits gained while using GLP-1s are quickly lost when a person stops taking the medication.

“The main message is that GLP-1 therapy behaves more like a long-term risk-reduction treatment than a short-term fix. The study reinforces a broader lesson in chronic disease management: benefits that accumulate slowly can be lost surprisingly fast when treatment is interrupted, so persistence and follow-up truly matter,” said Glatter.

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Millions Are Unaware of Cardiovascular Risks That Start Outside of the Heart

Healthline

Cardiovascular and metabolic health are intrinsically linked, with diabetes and kidney disease as major risk factors for heart disease. PER Images/Stocksy

A new report suggests that millions of U. S. adults are unaware they may have undiagnosed cardiovascular risk factors that begin outside the heart.

An estimated 90% of Americans may have a cluster of chronic health conditions known as cardiovascular-kidney-metabolic (CKM) syndrome, which can lead to cardiovascular disease.

A cardiologist explains the comorbidities of heart disease, diabetes, and kidney disease, and what you can do to protect your heart health.

When most people think of risk factors for cardiovascular disease, they likely consider their heart health.

Yet cardiovascular risks are also driven by other physiological processes that begin outside of the heart.

Metabolic health and cardiovascular health are intrinsically linked, with diabetes and kidney disease as significant risk factors for heart disease. Be that as it may, a 2025 survey shows that most people are unaware of this connection.

What's more, a recent statistics report by the American Heart Association (AHA) notes that as many as 1 in 4 adults with diabetes are undiagnosed in the United States. Additionally, 9 in 10 U. S. adults with chronic kidney disease are undiagnosed in the United States due to factors such as a lack of screening.

The comorbidity of diabetes, kidney disease, and heart disease means that having one of these conditions raises the risk of developing another due to shared risk factors, including:

high blood pressure

high cholesterol

high blood sugar

excess weight

reduced kidney function

The medical term describing the link between heart disease, diabetes, and kidney disease is cardiovascular-kidney-metabolic syndrome, or CKM syndrome, with upward of 90% of Americans unaware that they may have the disease.

If left untreated, complications from CKM syndrome can lead to disability and death from cardiovascular disease.

"We are encouraging people to become aware of the connection between conditions so they and their health care team can think about their overall health beyond individual conditions," Stacey E. Rosen, MD, volunteer president of the AHA, said in a statement.

"Understanding the connection helps you better prevent complications through lifestyle changes and appropriate treatment."

Other findings from the AHA statistics report highlight the prevalence of overlapping risk factors driving the prevalence of CKM syndrome in the United States:

around half of adults have high blood pressure

1 in 3 adults has high cholesterol

around 1 in 7 adults has kidney disease

more than half of adults have prediabetes or diabetes

over half of adults have a high waist circumference

"Each of these factors independently stresses the cardiovascular system," said Jack Wolfson, MD, a board certified cardiologist. "Together, they create a perfect storm."

Healthline spoke with Wolfson to learn more about how heart, metabolic, and kidney health are connected, and what people at risk for, or living with, CKM syndrome can do to minimize their cardiovascular risk.

This interview has been lightly edited and condensed.

How does metabolic syndrome cause heart problems?

Wolfson: Metabolic syndrome is not a single disease. It's a cluster of metabolic dysfunction that includes abdominal obesity, insulin resistance, elevated triglycerides, low HDL, and high blood pressure.

Insulin resistance drives chronically elevated insulin levels, which promote inflammation, endothelial dysfunction, and vascular stiffness.

High triglycerides and small LDL particles increase atherogenic burden. Elevated blood pressure damages the inner lining of arteries.

Visceral fat releases inflammatory cytokines that further impair vascular health.

Over time, this metabolic environment leads to accelerated plaque formation, impaired nitric oxide production, arterial stiffening, and left ventricular hypertrophy.

The heart is forced to pump against higher resistance while being supplied by progressively diseased vessels, whereas Metabolic syndrome is essentially early cardiovascular disease in motion.

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Bumps Inside Nose: Common Causes and Top Treatments

Greatist

Is it a zit, an ingrown hair, a polyp, or something else? Weird nose bumps have various causes, from harmless to time to call the doctor.

There are many possible causes for bumps inside your nose — some as harmless as a pimple, others that might need more attention. It's easy to imagine the worst, but the truth is usually less alarming.

If you've got a painful, annoying, or strange bump in your nose, keep reading to learn what it might be, whether it's worth worrying about, and how to deal with it.

Sergey Mironov/Getty Images

So, what's behind these bothersome nose bumps? Here are some common culprits:

Nasal vestibulitis

This is inflammation of the nasal vestibule, the area just inside your nostrils, often caused by bacterial infections. It can lead to painful, red bumps sensitive to the touch. Common triggers include frequent nose-picking, blowing too hard, or even plucking nose hairs.

Like on your face, the pores inside your nose can become clogged with oil, dirt, and bacteria, leading to pimples. These can be painful and may swell, especially in such a sensitive area.

Nasal polyps

Non-cancerous growths can develop in the nasal passages or sinus lining due to chronic inflammation, allergies, or infections. While they're usually painless, larger polyps can cause blockages, leading to breathing difficulties, a reduced sense of smell, or recurrent sinus infections.

Infected ingrown hairs

When you pluck or trim nose hairs, the hair can sometimes grow back into the skin instead of out, causing an ingrown hair. If this becomes infected, it can result in a painful bump inside the nose.

A bump can also form inside the nose due to trauma, such as a blow to the nose, or even from something as simple as aggressive nose-picking. The trauma can cause the tissue to swell or become bruised. In some cases, a hard bump may form as the tissue heals.

Fluid-filled sacs that can form in your nose, usually harmless but sometimes uncomfortable.

Cartilage issues

A hard bump inside the nose might be due to problems with the cartilage, such as a deviated septum or cartilage overgrowth.

Fibrous papules

Small, benign bumps that may appear on or in the nose are often mistaken for pimples but are fibrous tissue growths.

Wondering if that bump inside your nose is something serious? Here's what to look out for:

Pain or tenderness. Especially when touching or blowing your nose.

Redness or swelling. Around the area of the bump.

Discharge. You need to monitor the condition carefully to see if there's any pus or blood.

Difficulty breathing. If the bump is large enough, it might obstruct your airflow.

Persistent bumps. Anything that doesn't go away after a week should be checked out.

Ready to say goodbye to that pesky bump inside your nose? The proper treatment depends on what's causing it, but here are some ideas to ease the discomfort and promote healing:

Apply warm compresses. Apply gentle heat to the area to reduce swelling and promote drainage if it's a pimple or infection.

Use saline rinses. Keep your nasal passages clean with saline spray or a Neti pot.

Avoid nose-picking. Seriously, just don't. Not only is it urch, but it's a significant cause of trauma and infection.

Moisturize. Prevent dryness and cracking with a gentle, nose-safe moisturizer, especially in winter.

Leave ingrown hairs alone. Let them resolve independently or see a doctor if they get infected.

Additionally, if you're dealing with nasal polyps, a health-care professional might prescribe steroid sprays to shrink them and reduce inflammation, making breathing easier.

On the other hand, if a bacterial infection is the culprit, as in cases of nasal vestibulitis, your doc may recommend a topical antibiotic like mupirocin. This ointment is applied directly inside the nose to target the infection, reduce redness, and promote healing.

While most nose bumps aren't a big deal, some need more attention. Here's when to make that appointment:

Persistent or recurring bumps. If it's sticking around or keeps coming back, get it checked.

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What Is ‘Boy Kibble?’ The Simple High Protein Meal Trend That’s Going Viral

Healthline

Don’t let the name fool you. The “boy kibble” meal trend can be a surprisingly affordable, healthy high protein option—especially with a few simple recipe tweaks. Ivan Andrianov/Stocksy

“Boy kibble” is a food trend that’s been gaining popularity on social media, especially among Gen Z men.

Usually made up of ground beef and rice, the trend is being shared as an inexpensive, simple high protein meal to support muscle building.

Nutrition experts say the dish can provide important nutrients, but discourage eating it daily without modification.

Eating the same meal repeatedly may lead to nutritional gaps.

“Boy Kibble” is the latest viral meal trend that’s taking over social media platforms like TikTok and Instagram.

The trend has been described as the male counterpart to the 2023 “girl dinner” phenomenon, in which women shared casual, snack-style meals of small portions of various foods.

“Boy Kibble,” on the other hand, is exactly what it sounds like. Much like kibble for pets, it’s a basic, no-frills meal that typically consists of ground beef and white rice.

Across social media, fitness influencers and gym-focused creators say the dish helps them easily hit their daily protein targets without spending much time cooking or money on takeout.

Many claim they eat the meal several times a week, while some say it’s part of their everyday routine.

Unlike many viral social media diet trends, experts say “boy kibble” does have some benefits, especially when the recipe is modified.

However, it can also pose some risks if the basic recipe is eaten too often.

Nutritional benefits of “boy kibble”

Jennifer L House, MSc, RD, a dietitian with First Step Nutrition, says there’s a lot to like about the “boy kibble.”

“It’s cheaper than eating out, easy to make and batch cook, and nutrient-dense (especially if you add veggies),” she told Healthline.

“Almost 50% of our calories come from highly processed foods, so I support any kind of home cooking, no matter how basic.”

As for its nutritional plus points?

“Ground beef is a complete protein, which helps maintain muscle, repair tissue, and build antibodies. A 100g serving contains over 100% of our daily requirement for Vitamin B12, which is found [naturally] only in animal products,” House said.

“Beef contains other important minerals, too, such as zinc, which supports immunity, and iron, [which is necessary for hemoglobin], which transports oxygen in our bodies.”

However, other types of rice, such as brown rice, can offer greater nutritional value.

Healthline

She added that “boy Kibble” doesn’t have to be just for boys.

“The iron in beef is more important for women and growing children, who have higher iron needs. Many adult women are low in iron, which can lead to poor sleep, poor immunity, and lack of energy,” she said.

What about rice?

It’s important to note that not all rice is equal in nutrients. White rice, which is refined, has had its fiber and protein stripped out. This means that it behaves like a simple carb in the body.

However, other types of rice, such as brown rice, can offer greater nutritional value.

“The carbohydrates in rice complement ground beef as they provide the energy you need for your workout and to replenish the energy that was used during your workout,” Kezia Joy, registered dietitian nutritionist and medical advisor at Welzo, told Healthline.

“The two together may serve as a viable base for a post-workout or high-energy meal.”

The potential risks of eating basic “boy kibble”

While the basic “boy kibble” recipe has some nutritional merit, experts are raising concerns about how often some people claim to eat it.

“The basic ‘boy kibble’ meal includes meat and rice. If this is all someone ate, three times a day, they would be low in fiber, calcium, and some vitamins and minerals, such as Vitamin C and folate, as well as antioxidants found in fruits and vegetables,” House said.

She said she strongly discourages people from eating basic ‘boy kibble’ “for three meals a day, or even once a day, especially if it’s just meat and rice with no veggies added.”

Joy agreed, noting that while eating the same meal every day can make tracking calories or protein easier, “dietary variety is very important to ensure that you are getting all

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Rucking Benefits: Add Weight to Your Walk to Boost Fitness and Strength

Greatist

Wondering if rucking can be as effective as weightlifting? Spoiler alert: it definitely can!

Weightlifting isn't the only way to build muscle. Rucking, or walking with a weighted backpack, is also highly effective. It's easy to start, can be done anywhere, and helps build muscle, burn fat, and boost endurance.

Curious? Keep reading to discover why rucking might become your new favorite workout!

mixetto/Getty Images

Rucking is like taking your regular walk and cranking up the intensity with heavy cargo. It's as simple as strapping on a weighted backpack (aka rucksack) and hitting the pavement, the trail, or wherever you like to roam. Originally dreamed up by the military to toughen up soldiers, rucking, or a rucksack march has, er, marched into civilian life as a killer workout.

What makes it awesome? The extra weight transforms a casual stroll into a workout that means business. Plus, it's customizable — you can load up as much or as little weight as you want, making it perfect for any fitness level. And the best part is that you can ruck just about anywhere.

Rucking isn't just a walk in the park — it's a powerful workout that packs a punch in all the right places. Here's why rucking deserves a spot in your fitness routine :

Builds strength and endurance

Think of rucking as the ultimate multitasker for your body. It's like lifting weights but with cardio to keep your heart pumping.

As you lug that weighted backpack around, your legs, back, and core are putting in serious work, building strength, and endurance with every (weighted) step.

Burns calories and promotes weight loss

Want to torch calories without the treadmill monotony? Rucking's got you covered. The extra weight and steady movement combo turns your stroll into a calorie-burning machine. Perfect for shedding any additional pounds while keeping things interesting.

Improves posture and core stability

Carrying a weighted backpack helps your posture. Your core has to stay engaged to keep you balanced, which helps align your body and ward off injury.

If you've had a baby, rucking can be a game-changer, helping combat the postural shifts of pregnancy and hauling around kids. Plus, it builds the functional strength you

need for everyday life, like lifting groceries or wrangling toddlers.

Accessible and time-efficient

Rucking easily fits into a busy schedule. You don't need special equipment beyond a sturdy backpack or a gym membership, and you can do it while multitasking — think walking the dog in the morning or pushing a stroller. Plus, the whole family can join in, making it a great way to combine fitness with family time.

Starting rucking is easy, but here are some tips to help you get the most out of your rucks:

Start light. Begin with a challenging but manageable weight, around 10 to 15% of your body weight.

Choose the right backpack. A comfortable, sturdy bag with padded straps is essential to avoid strain and injury.

Focus on form. Keep your shoulders back, engage your core, and maintain a steady pace.

Increase gradually. As your strength and endurance improve, gradually increase the weight and distance.

Stay consistent. Regular rucking sessions will help you build strength and endurance over time.

Rucking is generally safe, but here's how to stay injury-free :

Listen to your body. Lighten your load or take a break if you feel pain or discomfort.

Warm up and cool down. Prepare your muscles for the workout and help them recover afterward.

Stay hydrated. Carry water with you, especially on longer rucks.

Wear supportive footwear. Good shoes can prevent foot and ankle injuries.

Be mindful of your surroundings. Pay attention to the terrain, and avoid areas that could be hazardous.

Does rucking build muscle?

Yes, rucking can help build muscle, especially in your legs, core, and back. The added weight provides resistance, similar to strength training.

What is the ideal weight for rucking?

Start with 10 to 15% of your body weight and adjust based on your comfort and fitness level. Heavier weights can affect your balance and stance, making injuries more likely. As you progress, you can gradually increase the weight.

Does rucking build traps?

Yes. Rucking forces your upper back and shoulder muscles

Improves posture and core stability Carrying a weighted backpack helps your posture.

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Notes on Linear Algebra for Polynomials

Eli Bendersky

We'll be working with the set $P_n(\mathbb{R})$, real polynomials of degree $\leq n$. Such polynomials can be expressed using $n+1$ scalar coefficients a_i as follows:

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

Vector space

The set $P_n(\mathbb{R})$, along with addition of polynomials and scalar multiplication form a vector space. As a proof, let's review how the vector space axioms are satisfied. We'll use $p(x)$, $q(x)$ and $r(x)$ as arbitrary polynomials from the set $P_n(\mathbb{R})$ for the demonstration. Similarly, a and b are arbitrary scalars in $\mathbb{R}$.

Associativity of vector addition :

$$p(x) + [q(x) + r(x)] = p(x) + q(x) + r(x) = [p(x) + q(x)] + r(x)$$

This is trivial because addition of polynomials is associative [1]. Commutativity is similarly trivial, for the same reason:

Commutativity of vector addition :

$$p(x) + q(x) = q(x) + p(x)$$

Identity element of vector addition :

The zero polynomial 0 serves as an identity element. $\forall p(x) \in P_n(\mathbb{R})$, we have $0 + p(x) = p(x)$.

Inverse element of vector addition :

For each $p(x)$, we can use $q(x) = -p(x)$ as the additive inverse, because $p(x) + q(x) = 0$.

Identity element of scalar multiplication

The scalar 1 serves as an identity element for scalar multiplication. For each $p(x)$, it's true that $1 \cdot p(x) = p(x)$.

Associativity of scalar multiplication :

For any two scalars a and b :

$$[a[b \cdot p(x)]] = ab \cdot p(x) = [ab] \cdot p(x)$$

Distributivity of scalar multiplication over vector addition :

For any $p(x)$, $q(x)$ and scalar a :

$$a \cdot [p(x) + q(x)] = a \cdot p(x) + a \cdot q(x)$$

Distributivity of scalar multiplication over scalar addition :

For any scalars a and b and polynomial $p(x)$:

$$[(a+b) \cdot p(x)] = a \cdot p(x) + b \cdot p(x)$$

Linear independence, span and basis

Since we've shown that polynomials in $P_n(\mathbb{R})$ form a vector space, we can now build additional linear algebraic definitions on top of that.

A set of k polynomials $p_k(x) \in P_n(\mathbb{R})$ is said to be linearly independent if

$$\sum_{i=1}^k a_i p_i(x) = 0$$

implies $a_i = 0 \quad \forall i$. In words, the only linear combination resulting in the zero vector is when all coefficients are 0 .

As an example, let's discuss the fundamental building blocks of polynomials in $P_n(\mathbb{R})$: the set $\{1, x, x^2, \dots, x^n\}$. These are linearly independent because:

$$a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n = 0$$

is true only for zero polynomial, in which all the coefficients $a_i = 0$. This comes from the very definition of polynomials. Moreover, this set spans the entire $P_n(\mathbb{R})$ because every polynomial can be (by definition) expressed as a linear combination of $\{1, x, x^2, \dots, x^n\}$.

Since we've shown these basic polynomials are linearly independent and span the entire vector space, they are a basis for the space. In fact, this set has a special name: the monomial basis (because a monomial is a polynomial with a single term).

Checking if an arbitrary set of polynomials is a basis

Suppose we have some set of polynomials, and we want to know if these form a basis for $P_n(\mathbb{R})$. How do we go about it?

The idea is using linear algebra the same way we do for any other vector space. Let's use a concrete example to demonstrate:

$$Q = \{1-x, x, 2x+x^2\}$$

Is the set Q a basis for $P_n(\mathbb{R})$? We'll start by checking whether the members of Q are linearly independent. Write:

$$a_0(1-x) + a_1 x + a_2(2x+x^2) = 0$$

By regrouping, we can turn this into:

$$a_0 + (a_1 - a_0 + 2a_2)x + a_2 x^2 = 0$$

For this to be true, the coefficient of each monomial has to be zero; mathematically:

$$\begin{aligned} a_0 &= 0 \\ a_1 - a_0 + 2a_2 &= 0 \\ a_2 &= 0 \end{aligned}$$

In matrix form:

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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Nicole ‘Snooki’ Polizzi Urges Cervical Cancer Screening After Diagnosis

Healthline

Nicole “Snooki” Polizzi has shared her recent cervical cancer diagnosis on her social media accounts. Image credit: Dimitrios Kambouris/Getty Images

Nicole “Snooki” Polizzi has shared her recent cervical cancer diagnosis on her social media.

She emphasized the importance of routine Pap smears and early detection.

Cervical cancer is largely preventable through the HPV vaccine.

Nicole “Snooki” Polizzi of “Jersey Shore” fame recently revealed that she has been diagnosed with stage 1 cervical cancer.

On February 20, Polizzi, 38, posted on TikTok about the cone biopsy she had after a routine pap smear.

“It came back stage 1 cervical cancer called adenocarcinoma,” the reality TV star said.

“Obviously not the news I’ve been hoping for, but also not the worst news just because they caught it so early. Thank freaking God!”

The American Cancer Society (ACS) estimates that around 13,290 new cases of cervical cancer will be diagnosed in 2026.

Cervical cancer is most frequently diagnosed between the ages of 35 and 64. Here’s what you need to know about getting screened.

Snooki encourages women to get Pap smears

In her TikTok videos, Polizzi also stressed the importance of all females going in to get their routine pap smears (cervical screenings).

“I’m 38 years old, and I’ve been struggling with abnormal pap smears for three or four years now, and now look at me,” she said.

“Instead of putting it off because I didn’t want to go, because I was hurt and scared, I just went and did it. And it was there, cancer is in there. But it’s stage 1, and it’s curable.”

She continued to tell people to get their appointments done. “Once you go to stage 2, then you have to do chemo... nobody wants to do that! It’s scary. So get your appointments done,” she encouraged.

Diana Pearre, MD, board certified gynecologic oncologist at The Roy and Patricia Disney Family Cancer Center at Providence Saint Joseph Medical Center in Burbank, CA, agreed.

“It is so important to get pap smears (cervical cancer screenings),” she told Healthline. “They allow us to screen women for HPV (the virus that causes cervical cancer) and identify cells that can become precancerous. In doing so, we can prevent many cases of cervical cancer before they transform to cancers.”

Polizzi said that she was being transferred to an oncologist and would undergo a PET scan to be sure the cancer has not spread to other parts of her body.

“After that, I’m gonna probably get the hysterectomy,” Polizzi added. She also noted that her doctor gave her the alternative of chemotherapy and radiation as treatment.

“Obviously, I think the smart choice here is the hysterectomy. I’ll still keep my ovaries, which is a good sign. But yeah, gotta get the cervix and uterus out. It all depends on the PET scan,” she said.

“I appreciate all of the love. Everything’s going to be fine. I’m going to tackle this and get it done,” Polizzi told her followers. “I gotta keep attacking this, and everything’s gonna be great.”

Cervical cancer largely preventable with HPV vaccine

While cervical cancer is common, it is also largely preventable.

According to the National Cancer Institute (NCI), 70% of cervical cancers are the result of two high-risk types of HPV.

Nearly all cases of cervical cancer are due to long-lasting and persistent HPV infections.

However, the HPV vaccine is a safe way to prevent the HPV infection and cervical cancer. The current recommendation is that anyone ages 11 to 26 should have the HPV vaccine.

“It is so important to get this vaccine. Giving children this vaccine (boys and girls alike) can prevent HPV related cancers (cervical, head and neck, vulvar, vaginal) before the onset of sexual debut. It can also help women who already have cervical dysplasia, lowering the risk of severe dysplasia recurrence,” Pearre said.

The vaccine dose schedule depends on your age when you receive it.

The vaccine is not recommended for everyone over 26, but you can speak with your healthcare professional to see if it is right for you.

“I recommend anyone ages 9 to 46 to consider getting the HPV vaccine if they have not done so,” Pearre said. “There are little to no side effects. It does not affect fertility, age at sexual debut, [or] menstrual patterns. It is one of the few vaccines (the other being the hepatitis vaccine) that can prevent cancer development.”

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Skin in the Game: Is Adapalene or Salicylic Acid Better for Acne?

Greatist

Adapalene and salicylic acid are both used to treat acne. We're going more than skin deep to discover the pros and cons of each.

If you have acne, you probably know more about skincare than most people. You might even impress friends with terms like "adapalene" and "salicylic acid."

But do you know what each treatment does for your skin and if one is better for acne? Let's find out!

Caroline Tompkins / Refinery29 for Getty Images

Salicylic acid is a keratolytic agent commonly found in over-the-counter creams, face washes, masks, and gels. While it's often used for acne, it can also help with dandruff, psoriasis, seborrheic dermatitis, calluses, and warts.

Adapalene, often known by its brand name Differin, is a topical retinoid. It comes in various forms, such as gels, cleansers, and creams. And just like salicylic acid, it's available over-the-counter without a prescription.

On a chemical level, adapalene and salicylic acid work in different ways to clear acne.

Salicylic acid breaks down a protein called keratin in the outer layer of the skin. Studies show this helps remove the buildup of dead skin cells. It also reduces inflammation in the skin, calming the redness that often comes with breakouts.

Adapalene, on the other hand, is a retinoid. And what, you may ask, is a retinoid? It's a derivative of vitamin A that attaches to receptors in the skin. Doing so prompts cell turnover to speed up, helping skin regenerate more quickly and lowering the likelihood of clogged pores.

So... what do these differences mean for acne? Both adapalene and salicylic acid can treat skin prone to breakouts (though it's not recommended to use them together since this can over-irritate your skin).

Which one is more potent may depend on your acne type. Research shows that adapalene was among the best treatments for comedonal acne (aka whiteheads and blackheads), whereas salicylic acid was less effective and more affordable.

Long story short: Your dermatologist can help determine which is the smarter choice for your type of acne.

Studies haven't directly compared adapalene and salicylic acid for blackheads, but both can help. Blackheads form from plugged hair follicles, and salicylic acid's exfoliating action can clear clogged pores. Adapalene increases cell turnover, which can also help remove blackheads.

Wrinkles: Adapalene vs. salicylic acid

Adapalene is a retinoid. Retinoids are well-established anti-aging agents because they stimulate new cell growth and promote collagen production. Some studies have also shown anti-aging benefits from salicylic acid.

While salicylic acid, due to its exfoliating properties, can help reduce wrinkles, adapalene is generally more effective for addressing aging-related concerns like wrinkles.

FAQs

What works better than adapalene?

Adapalene is an over-the-counter acne treatment — so prescription-strength options may work better. According to the American Academy of Dermatology, some of the most potent acne treatments include antibiotics and isotretinoin.

Is adapalene the best for acne?

Adapalene isn't a magic wand for acne. It may be an ideal choice for some people, but depending on your skin type and sensitivity level, salicylic acid, other OTC products, or even a prescription treatment might be your best bet.

Is adapalene stronger than benzoyl peroxide?

Research shows that adapalene can be just as strong for zapping zits as benzoyl peroxide (another potentially powerful weapon in the battle against acne). A 2024 study, for example, found that 0.1% topical adapalene was as safe and effective as a 4% benzoyl peroxide solution in people of color.

Is adapalene the strongest retinoid?

Though adapalene may be an effective acne treatment, it's not the pinnacle of retinoid strength. Of the four "generations" of retinoids, adapalene is considered third-generation. Fourth-generation retinoids are typically stronger and have higher specificity to retinoic acid receptors in the skin. For this reason, they're available only by prescription.

Feel free to experiment with adapalene and salicylic acid to see which works best for you — after all, it's your face! You might find one more effective for your skin. For expert advice, be sure to consult your dermatologist.

Which one is more potent may depend on your acne type.

Greatist

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Nearly 6 in 10 Women Will Develop Heart Disease, Stroke by 2050. What to Know

Healthline

Nearly 60% of U. S. women could have at least one type of cardiovascular disease by 2050. Ana Luz Crespi/Stocksy

Researchers say the heart health of American women will decline over the next 25 years if current trends continue.

A new AHA scientific statement projects that 60% of U. S. women will have at least one type of cardiovascular disease, 25% will have diabetes, and 60% will have obesity by 2050.

The authors say these trends are driven by rising rates of conditions such as high blood pressure and diabetes.

The American Heart Association (AHA) projects that the cardiovascular health of females in the United States will decline significantly over the next 25 years.

In a scientific statement published on February 25 in *Circulation*, the association reports that nearly 60% of U. S. women will have at least one type of cardiovascular disease by 2050.

The authors of the statement say the downward trend in women's heart health will cut across all age groups, all ethnicities, and all types of cardiovascular disease.

They report that the rise will be fueled by increases in health factors such as high blood pressure, diabetes, and obesity.

The authors also projected that rates of high cholesterol will decrease during the next quarter-century.

The scientific statements come on the heels of a recent study suggesting that women typically have less artery-clogging plaque than men, but still face higher rates of cardiovascular events like heart attacks.

The reports are part of the AHA's campaign to educate women about cardiovascular disease during American Heart Month.

"Cardiovascular disease is the leading cause of death for women and remains their number one health risk overall," Stacey Rosen, MD, volunteer president of the AHA, executive director of the Katz Institute for Women's Health, and senior vice president of women's health at Northwell Health in New York City, said in a statement.

Factors driving heart disease risk in women

Cardiovascular disease can affect women of all ages, with risk factors like high blood pressure driving the onset in younger females.

"We know the factors that contribute to heart disease and stroke begin early in life, even among young women and girls," Rosen said in a statement.

"The impact is even greater among those experiencing adverse social determinants of health such as poverty, low literacy, rural residence, and other psychosocial stressors. Identifying the types of trends outlined in this report is critical to making meaningful changes that can reverse this course," Rosen said.

The AHA's report noted that the expected increase in cardiovascular disease underscores the need for greater awareness and prevention strategies for U. S. women and medical professionals alike.

"We have done a great job decreasing deaths from big cardiovascular events like heart attacks and strokes, but these data suggest that we need to really refocus our efforts on health, wellness, and prevention," said first study author Karen Joynt Maddox, MD, a professor of medicine and public health and the co-director of the Center for Advancing Health Services, Policy & Economics Research at the Washington University School of Medicine in St. Louis, in a statement.

Jennifer Wong, MD, a cardiologist and medical director of Non-Invasive Cardiology at MemorialCare Heart and Vascular Institute at Orange Coast Medical Center in Fountain Valley, CA, wasn't involved in the scientific statement, but said it raises serious concerns.

"The overall rise in cardiovascular disease, despite reductions in hyperlipidemia and improvements in diet, physical activity, and smoking rates, is unexpected and concerning," Wong told Healthline.

"The rising prevalence of risk factors such as diabetes and hypertension, along with the marked increase in obesity among girls, is alarming," she said.

One third of younger women may develop cardiovascular disease

The authors of the AHA scientific statement said they expect several health issues to emerge by 2050 if current trends continue. Among their predictions:

Nearly 60% of U. S. women will have high blood pressure. That compares with fewer than 50% today.

More than 25% of U. S. women will have diabetes, compared to about 15% currently.

More than 60% of U. S. women will have obesity, compared to about 44% today.

The authors noted that older women will continue to have higher rates of cardiovascular disease. However, they said the rates among younger women will climb substantially. Among their projections:

Nearly a third of U. S. women ages 22 to 44 will have some type of cardiovascular disease by 2050, compared to 1 in 4

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7 Common Misdiagnoses for Hidradenitis Suppurativa (HS)

Greatist

Why this skin condition gets mistaken for others — and which ones.

Hidradenitis suppurativa (HS) is a skin condition that often gets caught up in a case of mistaken identity in a lineup of suspects. Specifically, people confuse HS for some more common conditions.

With their host of bumps, rashes, flakiness, or scales, skin conditions can sometimes be tricky to diagnose. That's because they often share common symptoms or have symptoms that overlap with or mimic those of another skin issue.

So, what exactly is HS? It's a chronic, progressive skin condition that typically appears as boil-like lumps in friction-prone and sweaty areas, where your skin can be sensitive. These lumps can form interconnected tracts underneath your skin (aka "tunneling") that may become irritated or even infected.

Knowing how the symptoms of HS compare with those of other inflammatory skin conditions can help you advocate for the correct diagnosis and care, which is crucial for staving off disease progression.

Here are 7 conditions that HS commonly gets mistaken for.

HS isn't rare — it affects up to 4% of the population. But by comparison, acne vulgaris affects more than 9% of people worldwide. Sometimes HS can be mistaken for acne simply because acne is more common. But these two conditions are quite different.

Acne typically shows up as papules, pustules, or nodules, which are commonly called zits or blemishes. Acne most often appears on your face but can also affect your back, chest, arms, and buttocks. It occurs when your sebaceous glands have a hypersensitivity to circulating androgens (like testosterone).

By comparison, HS most often appears in sweat- and friction-prone areas, such as your armpits or groin. It occurs when hair follicles become blocked — though more research is needed to learn what causes these blockages.

Acne conglobata (AC) is a rare but intense form of nodulocystic acne. With AC, the cysts and nodules connect under your skin, and this can lead to infections and scarring. Since AC features a "tunneling" effect that may appear similar to HS, many people confuse these two otherwise distinct conditions.

One key difference? HS often appears in areas that are subjected to friction from skin-on-skin contact or chafing. AC, on the other hand, can appear anywhere on your body. To further complicate things, many people with HS also have AC, and some research suggests a link between the two conditions.

A cyst is a fluid-filled sac. Cysts have many possible causes, including injuries and minor infections. They often go away on their own, but some may require treatment.

HS bumps can resemble cysts — but HS is an ongoing inflammatory condition.

An ingrown hair happens when a strand of hair anywhere on your body grows into your skin instead of out. Ingrown hairs commonly occur after shaving, waxing, or tweezing. After hair removal, a tiny new hair can curl inward, creating a bump (often called a razor bump).

Sometimes people confuse HS bumps for ingrown hairs — especially since HS often occurs in places where hair removal is common, such as the armpits and bikini area. Another confusing factor is that HS develops from blocked follicles. However, the cause of the blockage in HS is more complicated than an ingrown hair.

Folliculitis is inflammation of hair follicles, usually caused by a bacterial infection. The inflamed hair follicles show up as small bumps and can look like ingrown hairs.

People commonly mistake HS for folliculitis because the conditions often occur in the same skin areas.

Mild folliculitis tends to go away on its own with at-home treatment, though more severe forms may require antibiotics. HS, on the other hand, is a chronic condition.

A boil (also called a furuncle) is a bump filled with pus. The bump develops from a bacterial infection, often of a hair follicle. The infection causes a buildup of immune and skin cells, which leads to the formation of the bump. A carbuncle is a cluster of boils.

People often confuse HS with boils. However, boils are the result of an infection, and proper treatment typically resolves the issue. Meanwhile, the boil-like bumps of HS often require ongoing treatment.

Herpes is a common sexually transmitted infection (STI) caused by the herpes simplex virus. It shows up as painful sores or blisters.

Since herpes often affects the genital area and HS occurs around the groin or bikini line, people may confuse the two conditions. But while herpes can be spread through direct contact, HS is not contagious.

Other skin conditions can mimic HS, and vice versa. But a misdiagnosis can lead to an incorrect treatment that won't actually ease your symptoms or slow the condition's progression. This is true for other skin conditions as well.

That's why, if you have symptoms of any of the above conditions, you should talk with a knowledgeable dermatologist who can properly diagnose your concerns and find the right treatment plan.

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Mounjaro, Zepbound May Lower Risk of Eye Disease in People With Diabetes

Healthline

Tirzepatide may help lower the risk of diabetic retinopathy, a diabetes complication that can lead to vision loss. Douglas Sacha/Getty Images

Weight loss medications Mounjaro and Zepbound, which contain the active ingredient tirzepatide, may lower the risk of diabetic retinopathy.

This diabetes complication occurs when high blood sugar levels damage the blood vessels in the retina at the back of the eye.

Diabetes also raises the risk of other eye diseases, including cataracts and glaucoma.

A widely prescribed type of medication for the management of type 2 diabetes may also help reduce the risk of developing an eye condition that can lead to vision loss.

In a recent study published in *Ophthalmology*, researchers found that GLP-1 weight loss drugs like Mounjaro and Zepbound, which contain the active ingredient tirzepatide, may lower the risk of diabetic retinopathy in people with type 2 diabetes.

The researchers said that people with type 2 diabetes were less likely to develop new diabetic retinopathy or progress to more dangerous stages of the disease while taking tirzepatide medications.

“Based on findings obtained from a large database of patients across multiple clinical practices, those with diabetic retinopathy may be less concerned that taking tirzepatide is going to make their condition worse,” said Szilard Kiss, MD, an ophthalmologist at NewYork-Presbyterian/Weill Cornell Medical Center and lead author of the study, in a statement.

“The findings suggest they may have a reduced risk of requiring more eye treatments with lasers or injections, which are typically required when retinopathy becomes severe,” said Kiss, who is on the board of directors of Adverum Biotechnologies, which is owned by Eli Lilly, the manufacturer of Mounjaro and Zepbound.

In the new study, researchers pointed out that previous research had indicated that weight loss medications such as Ozempic and Wegovy could worsen diabetic retinopathy and associated conditions.

The previous study found that semaglutide medications produced some temporary deterioration of diabetic retinopathy in people with type 2 diabetes.

However, those researchers reported that the effects were short-term and that there was no long-term association between semaglutide and increased progression of diabetic retinopathy during the study’s 3-year period.

Tirzepatide lowers diabetic retinopathy risk

To reach their findings, the researchers in the new study analyzed electronic health records from 174,000 people across 70 health systems in the United States.

They focused on participants with type 2 diabetes with weight management issues, comparing people who started tirzepatide with similar patients who pursued only lifestyle interventions such as nutrition therapy or exercise counseling.

The researchers reported that after one year of treatment, the subjects taking tirzepatide had a lower risk of worsening diabetic retinopathy or related complications than those using lifestyle interventions.

As one example, researchers stated that incident mild non-proliferative diabetic retinopathy occurred in 0.49% of tirzepatide patients compared with 1.2% of the lifestyle intervention group.

Benjamin Bert, MD, an ophthalmologist at MemorialCare Orange Coast Medical Center in Fountain Valley, CA, said there may be a simple explanation for the effects of tirzepatide and semaglutide medications on diabetic retinopathy. Bert wasn’t involved in the study.

Bert explained that both types of medications help to suppress the body’s appetite and hunger responses by mimicking the body’s natural GLP-1 hormone. They also help manage blood sugar levels.

Adrian Au, MD, an ophthalmologist at UCLA Health in Los Angeles who wasn’t involved in the study, agreed that reduced blood sugar levels are a key factor in diabetic retinopathy. However, he noted there may be short-term risks.

“The concern with semaglutide largely relates to drops in blood sugar, which can temporarily worsen retinopathy in people who already have advanced disease,” Au told Healthline. “This phenomenon has been observed with intensive glucose control even before [the] popularization of GLP-1 medications.”

“It is likely more about how quickly glucose improves than about a harmful drug effect,” he added. “Over the long term, better blood sugar control is associated with improved retinal outcomes, so these short-term findings may not negatively affect long-term risk.”

As tirzepatide mimics the GLP-1 hormone, it leads to what the study authors described as “greater improvements in insulin sensitivity, weight loss and metabolic inflammation.”

“That’s encouraging, but the question is always how do these drugs perform over the long term?”

Au cautioned that the new study’s results are preliminary. “It’s observational data, so it shows association, not proof of cause and effect,” Au said.

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Ultimate Guide to 4A Hair: Care Tips, Styles, and Maintenance

Greatist

Understanding your 4a hair type is essential for achieving healthy, vibrant curls. Discover the best care tips for 4a hair, from moisturizing routines to protective styles, and keep your curls looking their best.

Understanding your hair type is key to achieving healthy, luscious curls. Hair types range from 1 to 4 — straight, wavy, curly, and coily — with type 4 being the curliest. Type 4A hair has a fine texture and loose curl pattern, forming springy coils that need special care to stay healthy and vibrant.

Knowing your hair type helps determine the best hair care routine. Caring for 4A hair is essential, whether you're dealing with dryness or breakage or want to define your curls.

Let's dive into the world of 4A hair and discover how to keep those curls popping.

Catherine Falls Commercial/Getty Images

Alright, curl friends, let's break it down. Hair type falls into four main categories :

Type 1: Straight

Type 2: Wavy

Type 3: Curly

Type 4: Coily

Now, Type 4 hair is where things get seriously curly and coily . Within this category, we've got subtypes A, B, and C, which refer to the width and density of your curls.

So, what's the deal with 4A hair? 4A hair is the queen of the fine, loose curl pattern . It forms tight, springy coils that are like tiny, perfect S-shapes. These curls are less dense than their 4B and 4C cousins, making them more defined and delicate. Think of 4A hair as the Goldilocks of coily hair — not too tight, not too loose, but just right.

Your hair is unique and gorgeous, but it requires a bit of extra TLC to keep it looking its best . Here's a straightforward guide to keeping your 4A hair moisturized, healthy, and thriving:

Moisturize regularly

4A hair tends to be dry as its coily structure makes it difficult for natural oils to travel down the hair shaft. Keeping it moisturized is essential , so incorporate a good leave-in conditioner into your daily shampooing routine. Use deep conditioning treatments weekly to replenish moisture and strengthen your curls.

Don't forget scalp care , too! For maximum hydration, look for products with ingredients like shea butter, coconut oil, and glycerin.

Gentle detangling

4A hair is prone to tangles, so detangle gently to avoid breakage. Use a wide-tooth comb or your fingers to work through knots carefully. Start from the ends and work your way up to the roots, and always detangle when your hair is wet and coated with conditioner to provide slip and minimize damage.

Avoid heat

Heat styling tools can cause significant damage to 4A hair, leading to dryness, breakage, and loss of curl pattern. Minimize the use of blow dryers, flat irons, and curling wands. If you use heat, always apply a heat protectant spray first to shield your hair from high temperatures.

Pro tip: To maintain the health of your curls, opt for air-drying or styles that don't require heat.

Protective styles

Protective styles like braids, twists, and buns can help reduce manipulation and minimize breakage . These styles tuck away your ends, protecting them from friction and environmental damage.

Satin or silk pillowcases

Cotton pillowcases cause friction , which leads to frizz and breakage, especially for 4A hair. Switching to satin or silk pillowcases instead helps mai

ntain your curl pattern and prevent tangles.

Regular trims

Regular trims are crucial for keeping 4A hair healthy and preventing split ends from traveling up the hair shaft. Aim to trim your hair every 8-12 weeks to remove damaged ends and keep your curls looking fresh and vibrant.

Is 4A hair curly?

Yes, 4A hair is curly and falls under the coily category. It features tight, springy curls that form an S-shape.

Is my hair 4A, 4B, 4C, or 3C?

To determine your hair type, look at the curl pattern, texture, and density. 4A hair has a loose curl pattern with fine, springy coils, while 4B and 4C hair have tighter curls. 3C hair is curly but not as tight as Type 4.

How often should you wash 4A hair?

Washing once a week or every two weeks is generally sufficient for 4A hair. Overwashing can strip the hair of its

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FDA Doesn't Endorse Leucovorin for Autism, OKs It for Rare Brain Disorder

Healthline

More research is still needed before leucovorin should be approved as an autism treatment. Westend61/Getty Images

Federal regulators have approved leucovorin for the treatment of cerebral folate deficiency, a rare neurological disorder.

The FDA approval, however, did not include the use of leucovorin as a treatment for autism.

The Trump administration has touted leucovorin as an autism treatment, but experts agree that more research is needed before leucovorin should be approved for this purpose.

A medication used primarily to help relieve the side effects of chemotherapy has been given the green light to be used as a treatment for a rare neurological disorder.

Officials at the Food and Drug Administration (FDA) announced on March 10 the approval of the drug leucovorin to help treat adults and children with cerebral folate deficiency. This rare condition is characterized by low levels of vitamin B9 in the brain.

However, the FDA announcement did not mention the use of leucovorin as a potential treatment for autism.

In September 2025, President Donald Trump and other administration officials suggested that the FDA had started a process to approve leucovorin as a treatment to manage symptoms in autistic people. However, there's no word yet on a timetable or process for such an approval.

Leucovorin is the first approved treatment for cerebral folate deficiency. The FDA decision is being praised as long overdue relief for individuals with that condition.

"Today's approval represents a significant milestone for patients living with cerebral folate transport deficiency due to the FOLR1 variant, a rare genetic condition that has had no FDA-approved treatment options until today," said FDA commissioner Marty Makary, MD, in the agency's announcement.

Leucovorin and cerebral folate deficiency

Leucovorin is a prescription medication used mainly to reduce the toxic side effects of chemotherapy agents such as methotrexate.

The medication works by helping to restore folic acid to healthy cells that have had that substance depleted by the effects of chemotherapy.

Leucovorin is considered to be effective as a "rescue agent" for people undergoing certain types of chemotherapy.

Cerebral folate deficiency is a condition that usually manifests in young children. It can cause:

developmental delays

seizures

movement abnormalities

It's estimated that 1 in 1 million people worldwide have cerebral folate deficiency, although the true prevalence is unknown. It's also been estimated that 38–70% of autistic children may have cerebral folate deficiency.

However, experts say those percentages may be inflated because much of the data has been drawn from FRAT tests, a blood exam that can be inaccurate.

While individuals with cerebral folate deficiency may have a higher risk of autism, but autistic people don't necessarily have a higher risk of cerebral folate deficiency.

Alycia Halladay, MD, the chief science officer for the Autism Science Foundation, said she's glad that people with cerebral folate deficiency finally have a treatment.

Healthline

Alycia Halladay, MD, the chief science officer for the Autism Science Foundation, said she's glad that people with cerebral folate deficiency finally have a treatment.

"This will probably help them. The mechanism is there," she told Healthline. "I feel any relief would be helpful."

Antonio Hardan, MD, a professor of psychiatry and behavioral sciences at Stanford University, said that early intervention with leucovorin for cerebral folate deficiency is key.

"Case reports [support] complete clinical and radiological recovery when treatment is provided before the age of 2 years," Hardan told Healthline.

While early intervention in autism is also crucial, it's too soon to recommend leucovorin as a treatment.

Halladay expressed relief that leucovorin wasn't approved as an autism treatment because there isn't yet a scientific justification for this particular use.

"I'm glad the FDA used a scientific approach and used rigor to make this decision," she said.

Hardan echoed this sentiment. "There is some evidence from small randomized controlled trials supporting the use of leucovorin for the treatment of autism. However, these studies have several design limitations, including small sample sizes and non-rigorous inclusion/exclusion criteria," he said.

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noom med epm | GLP-1RX Program

Greatist

Why Noom's GLP-1 Rx Program is the 'Hype Friend' You Didn't Know You Needed

Clinically-Backed, Customized to You

Psychology-based approach + prescription GLP-1s = lose weight and keep it off

Access to GLP-1s starting at \$99 if you qualify

Everything revolves around what you need

Full access to licensed clinicians, behavior change coaches, and community groups

Your Friend is a Proven and Trusted Program

Most weight loss programs give you a prescription or a plan and send you on your way. Not Noom.

Noom Med combines the clinician-prescribed GLP-1 medication with the psychology-based approach that's already helped millions.

Noom's primary goal is to empower everyone everywhere to live better, longer. All of its programs, courses, efforts, clinicians, and team of specialists are designed to help you reach your goal.

All while you have a team of specialists to guide you through the process.

Take A Hand and Jump In

The process shows and affirms that there is more than one way to achieve your health goals: The way that works best for you. To get started:

Take the quiz

Connect with a qualified clinician

If you qualify, start a customized treatment plan with medication

Have ongoing access to care and personalized support

Lose 48% more weight with Noom Med than medication alone 1

97% doctors recommend pairing GLP-1s with lifestyle changes 2

9 in 10 doctors recommend combining GLP-1s with nutritional support. That's why Noom is more than meds and helps you lose weight and keep it off 2

The Companion, GLP-1 Companion

If you qualify for GLP-1 medications, you will have access to Noom's GLP-1 Companion for no added cost.

GLP-1 Companion is a tool and friend that provides dedicated support from the beginning of your weight-loss journey through long after to help you maintain your weight.

It offers resources and support to guide you on everything related to GLP-1 use to ensure you're getting the most out of your medication, including:

Smartdose Guidance to support clinician-led treatment

AI Nutrition Guidance helps create personalized plans

Muscle Defense workouts tailored to you

An AI Body Scan that can help track body composition

Other tips on movement, side effects, and more

"I had never known the psychology behind the way we eat. This was all new information for me to learn. I had a slow start, but I didn't give up. I decided to give Wegovy a try a few months back... my weight loss has really improved since that time. My mindset about food and eating has changed a lot since June of 2022.

I'm not done yet. I'm still on my journey!"

— Jennifer L., Noom Med User | Age 53

What's included in Noom Med?

Noom Med combines personalized medication and its signature psychological support to help make healthy habits last and make transformational weight loss possible.

It also gives you access to 1:1 messaging with expert weight loss clinicians, prescriptions and medications (where clinically appropriate), ongoing medication management, community support, and psychology-backed content to make lasting changes.

How can I get access to Noom's GLP-1 Companion

If you qualify for GLP-1 medication, you have access to Noom's GLP-1 Companion, a digital health experience that supports people taking GLP-1s. It helps navigate treatment, symptoms, side effects, nutrition, and more.

How much do GLP-1s cost with Noom?

Noom offers GLP-1 Rx programs with direct access to GLP-1s starting at \$99, regardless of insurance coverage, if clinically appropriate and prescribed.

Noom Med includes brand-name medications like Ozempic® and Zepbound®, starting at \$69 plus the out-of-pocket costs for the medication itself, depending on your insurance coverage.

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Why Does My Side Hurt When I Run? Causes and Prevention Tips

Greatist

Experts don't know for sure but think side cramps during exercise may be caused by diaphragm issues, poor posture, or nerve irritation from your foot striking the ground.

Side stitch, stomach cramp, side ache — whatever you wanna call it, that mid-run pain in your tum can seriously *cramp* your runner's high.

This pesky prob that plagues runners and athletes all over has a scientific name: Exercise-related transient abdominal pain (ETAP) . Though experts don't know exactly what causes it, they have some ideas and tips about how to stop it.

Before you double-over during your next marathon , here's what to know.

KarraStock/Getty Images

That fateful stab in your side is most likely to happen during exercise where your upper bod says tight and upright for an extended period, such as when you run , jog, bike, hike, play soccer, etc.

This common prob has been around for ages — the first historical mention is from ancient Roman philosopher Pliny the Elder . Yet, scientists still haven't reached a consensus on its exact cause. We know that it affects up to 70% of runners each year.

It's also more likely to occur on the right side.

A stiff spine or poor posture (according to older research from 2010) could contribute to side stitches – but TBH, the pros don't know for sure.

According to a 2022 review , researchers also think a side stitch might be caused by your right foot hitting the ground while you breathe out, irritating a phrenic nerve. Researchers think this, in turn, may bother the lining of your abdomen (particularly on the lower right side). Since most people are right-foot dominant, this is one possible explanation.

Experts also theorize that it could be triggered by the boost of blood flow around the diaphragm when running .

Furthermore, fueling up before a big race is excellent — but eating or drinking large quantities right before running may boost your risk of getting a side stitch, anecdotal reports suggest.

When a side stitch strikes, don't stress – there are steps you can take, including:

Slowing down , which should help lend some immediate relief. When the pain subsides, you can slowly build your pace back up.

Stretch it out by raising your arms overhead, leaning to the opposite side of the pain, and breathing deeply. That may open up your abdomen and offer some relief.

Apply gentle pressure by pressing your palm into the cramp while breathing deeply.

To stop a side stitch before it begins, here are some ideas FTW:

Practice mindful workout breathing. According to a 2022 review , slower breathing during exercise might “trick” your bod into thinking the exercise is easier, increasing pleasant feelings and reducing stress. While the study doesn't mention side cramps, it could be worth a try since stitches may be linked to blood flow surrounding the diaphragm.

Avoid big meals or beverages a couple of hours before you run. Though experts don't know for sure, too much food or liquid in your belly could potentially put extra pressure on your precious organs, leading to a cramp.

Work on your posture . Proper posture can improve breath flow and running performance. Research from 2014 suggests building up your abs could help prevent side stitches.

Stay hydrated to stay healthy in general and potentially prevent side stitches.

Start slow and work your way up. When working out, starting small and gradually upping your challenge helps reduce the risk of injury. In the same way, gradually increasing your speed rather than diving straight into a sprint may also help reduce the risk of a side stitch.

When to see a doctor

Side stitches are usually harmless. But it's always a good idea to visit a pro if you experience any of the following probs:

Pain under your breastbone (higher upper than a side stitch) accompanied by difficulty catching your breath could indicate a heart or lung issue.

Persistent pain after running could indicate a chronic condition.

Severe or debilitating stomach pain or cramps could also signal a more serious condition.

Side stitches are a common issue and not typically the reason for concern. Though pros don't know what causes them, they think issues with breathing, posture, or nerves may play a role.

Consider working on your running form or breathing to reduce the risk of experiencing a side stitch. Staying hydrated and avoiding huge meals pre-run may also help. Consider working with a physical therapist or personal trainer if your issues persist.

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28 Best Brunch Recipes For A Crowd

Lindsay · Pinch of Yum

I love a good brunch moment! Whether it's for something special like Easter or a baby shower, or just a simple Saturday morning with friends. Breakfast food is some of the best food (sweet, savory, cheesy, carby!) and I love to see a table full of people gathered around it. ♥

These are the brunch recipes I put on the menu when I'm hosting a group – some can be thrown together and served out of one pan for ease, others can be assembled however your guests see fit (LOVE this breakfast sandwich for that), and all of them are simple, easy, and super delicious.

One of my favorite menu combos: this egg and croissant bake , this simple green salad , and this carrot cake coffee cake . The ultimate brunch plate!

Here's to brunch! Here's to feeding all our favorite people!

class="tasty-roundups-description">Biscuits and Gravy Egg Bake! YES! a casserole-style dish of fluffy eggs and cheddar baked up with little bites of tender flaky biscuit pieces, and buried under a blanket of creamy green chile sausage gravy.

class="tasty-roundups-description">Crunchy, light, bright, and ultra-fresh. This French-inspired bistro salad is an absolute star as a side for nearly any dinner. Tender greens, fresh herbs, pickled onions, and a delicate vinaigrette on top.

class="tasty-roundups-description">Sweet Cinnamon Rolls! Made with pillowy brioche dough and cinnamon sugar tucked into every nook and cranny. This makes a BIG batch of dough that you can divide and save for later...or make extra all for yourself!

One of my most important brunch rules is that it must include multiple drinks – a glass of orange juice, a cup of coffee, and maybe something special, too? Here are some ideas for drinks to serve with brunch, many of which can be scaled up for a crowd!

The post 28 Best Brunch Recipes For A Crowd appeared first on Pinch of Yum .

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standard-article(title: [The Trip That Changed Me: How Running the World's Biggest Marathons Pushed AnneMette Bontaites's Limits], author: [AnneMette Bontaites], source-name: [Atlas Obscura], images: (), paragraphs: ([For AnneMette Bontaites, running the New York City Marathon was supposed to be a one-time experience. A personal challenge, a bucket-list item, and then back to normal life], [But somewhere between the starting line and the finish, plans changed.], [The decision to run that first marathon

standard-article(title: [The Buried Church Beneath Musée Rude in Dijon, France], author: [Atlas Obscura], source-name: [Atlas Obscura], images: (), paragraphs: ([Beneath the elegant calm of the museum lies something far older and far less orderly: the archaeological remains of an 11th-century church that refuses to fully disappear. The crypt is not polished or prettified. Instead, it reveals raw foundations, fractured columns, and worn stone outlines that trace the ghostly footprint of the original structure.], [This

was almost a whim. Bontaites, a Denmark native who now lives in Boston, had run two half marathons and joked to her best friend, “Well, two halves make a whole. I’ve done a marathon.”], [The friend, also a runner, begged to differ on that distance math and made her an offer: She’d fly to New York from the pair’s home city of Copenhagen and they’d run Bontaites’s first marathon together. By mile 18, the friends had made a pact to run the Copenhagen marathon together, too.], [Soon the races began to stack up: After Copenhagen came Paris, the Marine Corps Marathon, and eventually the prestigious Abbott World Marathon Majors series, which includes seven races in Berlin, Boston, Chicago, London, New York, Sydney and Tokyo.], [“Before you actually run one, I think in the back of your mind, most people [think], ‘There is no way I could ever run a marathon.’ And then when you do, it’s this really incredible fulfilling feeling of, ‘Wow, I really did this.’”], [And the discipline and mental grit required to train and run those races also reshaped how Bontaites approaches challenges far beyond the course.], [“It helps when you actually break down the 26.2 miles into five-mile increments because then it becomes less daunting mentally,” she says. “Itake that to work with me.”], [The lesson, she discovered, wasn’t just about running—it was about reframing overwhelming goals into manageable steps.], [Bontaites completed the Abbott series in August 2025 when she ran the Sydney Marathon and considered retiring, but then Athens – birthplace of the marathon – came calling. That race was in November.], [“I thought, ‘Let’s end where it all began,’” she said.], [Follow AnneMette’s travels through the Atlas here.], [Do you have a trip that changed you? Fill out our form with your name, email, a brief description of the moment that shifted something for you, and a few photos. We may feature your story in future content — and you may even be invited to be interviewed for the series. And read more from our “The Trip that Changed Me” series here.], [Loading...],), insert-map: (:), word-count: 395, edited-for-length: false, debug-mode: false,)

underground space once belonged to the Church of Saint-Étienne, one of the earliest religious centers in Dijon . Time, revolutions, and urban reinvention dismantled the church above, but its bones endured.], [Rough stone arches curve overhead like ribs. The air is cool and still, carrying the quiet dignity of something that has outlived its purpose yet refuses to vanish. Informational panels help decode the fragments, but the true magic lies in the unfinished feeling. Nothing here tries to impress. It simply persists.], [The result is a rare encounter with a building in reverse, not restored to glory, but preserved mid-disappearance. It’s less a monument than a memory made visible.],), insert-map: (:), word-count: 164, edited-for-length: false, debug-mode: false,)

‘And then when you do, it’s this really incredible fulfilling feeling of, ‘Wow, I really did this.

AnneMette Bontaites

standard-article(title: [This Intrepid 19th-Century Reporter Refused to Accept the Unacceptable], author: [Daniel McDermon], source-name: [Atlas Obscura], images: (), paragraphs: ([This article comes from Atlas Obscura’s Places newsletter. Subscribe or manage your subscription here .], [The enterprising 19th-century journalist Nellie Bly didn’t just write stories—she stepped into danger to force readers to see things that they might prefer to ignore. Bly went undercover in the 1880s to expose the asylum system, in which women (with or without mental illness) were often abused or neglected. More than a century later, Bly’s work was honored with a monument near the site of the asylum she investigated, on New York’s Roosevelt Island.], [“The Girl Puzzle,” named after one of the journalist’s early works, includes five monumental faces of women, one of them Bly’s, along with four spheres of mirror-polished steel. It’s a reminder on this International Women’s Day that progress

standard-article(title: [A Week In The Life, Vol 12.], author: [Jessica], source-name: [How Sweet Eats], images: (), paragraphs: ([We had a fun little week!], [These little irish babies.], [Um we woke up to lots of snow! The kids even had a delay. What the heck.], [These two working hard on their brackets.], [OG BFFs.], [The leprechaun dyed our milk green.], [But don’t worry he doesn’t bring presents. Ugh.], [Cutie.], [Still on my brown sugar espresso kick.], [This outfit that is so perfectly 80s.], [Jordan is so obsessed with these powdered white donuts.], [And then it was our school’s drama club play!!], [Max and his best friend made this elephant and they were so proud of it!], [All of his cousins came to see him the first day!], [He was so into it and so proud, it was adorable.], [Then we had a fun cousins day!], [Baby Lucy is so edible.], [I threw together these buffalo chicken wraps which were SO good.], [Then we had the final night of The Lion King!], [Leading the pack to the circle of life.], [View

has often depended on women who refused to accept the unacceptable.], [In Manchester, England, a sculpture of the suffragette Emmeline Pankhurst is frozen mid-speech , rallying an invisible crowd toward votes and visibility. The Musée de La Femme in Marrakesh spotlights Moroccan women’s creative and civic power . In Senegal, the Henriette Bathily Women’s Museum stands as a tribute to women’s cultural contributions . And La Casa Azul in Mexico City preserves Frida Kahlo’s intensely personal world .], [Female visual artists have long had to struggle, not just to have their work widely seen, but to create at all. But women have made art as long as there’s been art. Many persevered, both through the strength of their work and the force of will, and in celebration of Women’s History Month, we want to highlight some of our favorite places where you can see these contributions in person. SEE THE FULL LIST], [The Music Box Theatre in Chicago is a historic 1929 cinema showing independent and classic films in an atmospheric setting.], [A plane wing with the Mechanics’ Creed written on it is a roadside art installation and monument on a remote stretch of highway in Iceland.], [An unexpected mix of medieval buildings and modern art enlivens this street in Morlaix, France.], [Visitors were drawn to mental institutions out of curiosity and compassion, but they weren’t seeing the full picture—until Nellie Bly revealed it.], [The Undercover Woman Who Changed Asylum Tourism Forever],), insert-map: (:), word-count: 385, edited-for-length: false, debug-mode: false,)

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ANALYSIS

from the back.], [Emilia’s dream.], [Finished the weekend off with our baby bff birthday!], [A painting party!], [These two.], [This cupcake cake was so cute.], [And the birthday girl is the sweetest!], [The post A Week In The Life, Vol 12. appeared first on How Sweet Eats .],), insert-map: (:), word-count: 210, edited-for-length: false, debug-mode: false,)

Maultaschen

Atlas Obscura

The origins of Germany's Maultaschen are deliciously devious. Legend has it that, in the late Middle Ages, a lay brother named Jakob invented the stuffed pasta dumplings at the Maulbronn Monastery, a UNESCO World Heritage site founded in 1147 by Cistercian monks in southwest Germany.

One direct translation of Maultaschen is “mouth pockets,” though “Maul” could just as easily refer to Maulbronn. Maultaschen are usually square dumplings (though sometimes they're rolled) and can be fried in a pan or served in broth. Commonly described as Germany's version of Italian ravioli, they allegedly emerged as a way to use up an unexpected bounty of meat that Brother Jakob stumbled upon in the forest outside the monastery walls.

The twist? Although they abhorred waste, these monks weren't allowed to eat the meat of four-legged animals, especially during the Catholic fasting period of Lent in the spring. So Brother Jakob minced the meat with herbs and onions and wrapped everything inside pasta dough, hiding the forbidden flesh from the eyes of his fellow monks—and even from the eyes of God.

In Swabia, the region encompassing much of Baden-Württemberg and part of Bavaria where Maultaschen originated, one of the colloquial names for the food references this deception directly: Herrgottsbescheißerle means “little God-cheaters.”

Everyone in Swabia has their version of the legend with more or less embellishment. Ludwig Nestler holds a master's degree in heritage conservation and works for the State Palaces and Gardens of Baden-Württemberg, a government organization that oversees monuments like Maulbronn Monastery. His version of the tale includes a sack of stolen meat dropped in the woods by a fleeing thief, which inspires Brother Jakob's trickery in the kitchen. But he acknowledges that there's no undisputed “historically correct version” of how Maultaschen came to be. Similarly, everyone in Swabia has their own Maultaschen recipe, with unique ingredients for the minced filling, called Brät .

“Traditionally the Brät is made from pork mixed with herbs, onions, and occasionally bread crumbs for texture and stability,” says Nestler. Swabia, however, “was a rather poor region with limited amounts of meat due to rather unfertile land, so being adaptive and innovative has always been a part of the people's nature.” As Maultaschen became popular, fish and seasonal vegetables like spinach, carrots, beets, and mushrooms became common inclusions.

Today, the European Union ties Maultaschen to Swabia with a Protected Geographical Indication , which lists required ingredients the authentic product should feature, but even the necessary inclusions are pretty loose, such as “pork and/or beef and/or veal” for meat Brät and “typical regional

vegetables” for meat-free Brät. It speaks to the way the dumplings developed as subsistence food, used to stretch leftovers and reduce food waste.

Today, Germans throughout the country enjoy Maultaschen in dozens of flavors in all seasons thanks to grocery stores that stock packaged varieties made by companies like Ditzingen-based Bürger, whose mascot, Erwin , is a Maultasche (the singular form of the plural Maultaschen).

But the dumplings remain most popular in southern Germany. Maulbronn Monastery offers a special tour that pairs Maultaschen with wine from the monastery's vineyards. And many locals, including Nestler's family, still make them from scratch on special occasions—even during Lent, when meat might otherwise be off the menu. There's no telling if it's a fraud good enough to fool God, but it's worth a shot.

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6 Signs Your Skin Needs Medical Attention, Stat

Greatist

These symptoms may signal an underlying condition and the need to see a dermatologist.

Your skin is your most visible organ, so it's easy to notice when something seems off. It's normal to experience some skin issues from time to time, like itchiness, breakouts, or ingrown hairs. But when these skin symptoms stick around, it could be a sign that something more serious is going on.

Many of these skin issues get better with the right at-home remedies or changes in your skin care routine. For example, if dryness pops up when Old Man Winter comes along, slathering on a more intense moisturizer may help.

But if skin symptoms don't go away with a little self-care — or they return frequently — a chronic skin condition might be the culprit. However, just because you have a lasting symptom or two doesn't necessarily mean you have a chronic health issue.

Here are the symptoms you should watch for and what they might mean. And if they don't improve, consider checking in with a dermatologist for a proper diagnosis and a plan of action.

If you wish you had cat claws to satisfy your skin itch, you could be dealing with eczema. Eczema is an umbrella term for several conditions that impact the skin, with atopic dermatitis being the most common.

With most forms of eczema, your skin may show signs of inflammation, such as:

itchiness

red, gray, brown, or purple skin

a rash

intense dryness

These symptoms may flare up occasionally and then go into periods of remission, where your skin isn't as irritated.

Allergies can also lead to itchiness. For example, if you're allergic to a skin care ingredient, a fabric type, a pet, or another allergen, your skin might react as a way to tell you to avoid this trigger.

Ongoing or regular periods of itchiness are a reason to see a dermatologist.

A rash can affect the skin in various ways, including:

red, gray, brown, or purple patches

itchiness

raised bumps

hives (urticaria)

A one-off or temporary rash might result from friction or chafing from a new outfit item. It could also be a heat rash when sweat gets trapped in skinfolds or bodily creases.

A persistent rash could be a sign of eczema or allergies. However, different types of rashes can be a sign of other things. For example, a butterfly-shaped (malar) rash across the face is a common sign of an autoimmune disease known as lupus.

If you frequently experience skin rashes, talk with a dermatologist to help you get to the bottom of the issue.

Dry skin can be just that — dry skin. Common causes include:

cold weather

dry weather

dry indoor heat

super-hot baths or showers

bathing too frequently

chlorine exposure from swimming

irritation from ingredients in skin care or soaps

sun exposure

But if your skin is so dry it makes you want to bathe in a vat of thick moisturizer, it could be a sign of psoriasis.

Psoriasis is an inflammatory skin condition potentially caused by an overactive immune system. It can lead to dry, cracked, or scaling skin. In fact, psoriasis often involves a buildup of scales or plaques on areas like the elbows or knees — though they can occur anywhere.

Psoriasis and eczema are different conditions, but they have overlapping symptoms that mimic each other. So if you experience these symptoms, it's important to see a dermatologist to identify and treat the underlying cause.

Changes in skin color — including red, brown, gray, or purple patches of skin — can be a sign of various skin issues, including eczema, psoriasis, or allergies. But it can also be a sign of rosacea.

Rosacea typically involves persistent redness that looks like a sunburn or blushing. It may also lead to skin thickening.

The good news? Rosacea is highly treatable. So working with a dermatologist can help you address it.

No one is immune from developing the occasional pimple or ingrown hair. But sometimes, skin bumps can be a sign that something else is going on.

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Notes on Lagrange Interpolating Polynomials

Eli Bendersky

Polynomial interpolation is a method of finding a polynomial function that fits a given set of data perfectly. More concretely, suppose we have a set of $n+1$ distinct points $[1]$:

$$[(x_0, y_0), (x_1, y_1), (x_2, y_2) \cdots (x_n, y_n)]$$

And we want to find the polynomial coefficients $\{a_0 \cdots a_n\}$ such that:

$$[p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n]$$

Fits all our points; that is $p(x_0) = y_0$, $p(x_1) = y_1$ etc.

This post discusses a common approach to solving this problem, and also shows why such a polynomial exists and is unique.

Showing existence using linear algebra

When we assign all points (x_i, y_i) into the generic polynomial $p(x)$, we get:

$$\begin{aligned} p(x_0) &= a_0 + a_1 x_0 + a_2 x_0^2 + \cdots + a_n x_0^n = y_0 \\ p(x_1) &= a_0 + a_1 x_1 + a_2 x_1^2 + \cdots + a_n x_1^n = y_1 \\ p(x_2) &= a_0 + a_1 x_2 + a_2 x_2^2 + \cdots + a_n x_2^n = y_2 \\ &\vdots \\ p(x_n) &= a_0 + a_1 x_n + a_2 x_n^2 + \cdots + a_n x_n^n = y_n \end{aligned}$$

We want to solve for the coefficients a_i . This is a linear system of equations that can be represented by the following matrix equation:

$$\begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

The matrix on the left is called the Vandermonde matrix . This matrix is known to be invertible (see Appendix for a proof); therefore, this system of equations has a single solution that can be calculated by inverting the matrix.

In practice, however, the Vandermonde matrix is often numerically ill-conditioned, so inverting it isn't the best way to calculate exact polynomial coefficients. Several better methods exist.

Lagrange interpolation polynomials emerge from a simple, yet powerful idea. Let's define the Lagrange basis functions $l_i(x)$ ($i \in [0, n]$) as follows, given our points (x_i, y_i) :

$$[l_i(x) = \begin{cases} 1 & \text{if } x = x_i \\ 0 & \text{if } x = x_j \text{ for } j \neq i \end{cases}]$$

In words, $l_i(x)$ is constrained to 1 at $x = x_i$ and to 0 at all other x_j . We don't care about its value at any other point.

The linear combination:

$$[p(x) = \sum_{i=0}^n y_i l_i(x)]$$

is then a valid interpolating polynomial for our set of $n+1$ points, because it's equal to y_i at each x_i (take a moment to convince yourself this is true).

How do we find $l_i(x)$? The key insight comes from studying the following function:

$$[l'_i(x) = (x-x_0) \cdots (x-x_{i-1}) \cdots (x-x_{i+1}) \cdots (x-x_n) = \prod_{\substack{0 \leq j \leq n \\ j \neq i}} (x-x_j)]$$

This function has terms $(x-x_j)$ for all $j \neq i$. It should be easy to see that $l'_i(x)$ is 0 at all x_j when $j \neq i$.

What about its value at $x = x_i$, though? We can just assign into $l'_i(x)$ to get:

$$[l'_i(x_i) = \prod_{\substack{0 \leq j \leq n \\ j \neq i}} (x_i - x_j)]$$

And then normalize $l'_i(x)$, dividing it by this (constant) value. We get the Lagrange basis function $l_i(x)$:

$$[l_i(x) = \frac{l'_i(x)}{l'_i(x_i)} = \prod_{\substack{0 \leq j \leq n \\ j \neq i}} \frac{(x-x_j)}{(x_i-x_j)}]$$

Let's use a concrete example to visualize this. Suppose we have the following set of points we want to interpolate: (1,4), (2,2), (3,3). We can calculate $l'_0(x)$, $l'_1(x)$ and $l'_2(x)$, and get the following:

Note where each $l'_i(x)$ intersects the axis. These functions have the right values at all x_j ($j \neq i$). If we normalize them to obtain $l_i(x)$, we get these functions:

Note that each polynomial is 1 at the appropriate x_i and 0 at all the other x_j ($j \neq i$), as required.

With these $l_i(x)$, we can now plot the interpolating polynomial $p(x) = \sum_{i=0}^n y_i l_i(x)$, which fits our set of input points:

Polynomial degree and uniqueness

We've just seen that the linear combination of Lagrange basis functions:

$$[p(x) = \sum_{i=0}^n y_i l_i(x)]$$

is a valid interpolating polynomial for a set of $n+1$ distinct points (x_i, y_i) . What is its degree?

Since the degree of each $l_i(x)$ is n , then the degree of $p(x)$ is at most n . We've just derived the first part of the Polynomial interpolation theorem :

Polynomial interpolation theorem : for any $n+1$ data points $(x_0, y_0), (x_1, y_1) \cdots (x_n, y_n) \in \mathbb{R}^2$

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About Me in Dating App: Write the Perfect Profile Bio

Greatist

Crafting a sizzling “About Me” section can transform your dating profile from drab to delicious. Find out how.

You’ve swiped right and gotten matches, but it’s not working. It might be your profile that’s holding you back. With just seconds to make an impression, your “About Me” section is crucial. Think of it as your personal highlight reel—enticing and leaving them wanting more. If it’s not working its magic, you could be missing out.

Read on to learn how to turn that blank box into a captivating snapshot of your fabulous self.

Sophie Mayanne/Getty Images

Your dating profile is your chance to show potential matches why you’re a catch. But what should you include? Since everyone is unique, think about what you bring to the table and what you’re looking for. Let’s break it down.

Showcase your personality: Are you a quirky adventurer, a bookworm, or a foodie always on the hunt for the best taco in town? Share your passions and what makes you tick. Be authentic — people can spot a fake from a mile away.

Be specific: Avoid generic statements like “I love to have fun” or “I enjoy hanging out.” Instead, get specific. Do you love hiking? Mention your fave trail. Are you a movie buff? Share your top films. Details increase relatability and spark conversations.

Use humor: Humor is desirable ! A witty one-liner or a playful anecdote can make your profile memorable. Just keep it light and fun — avoid heavy sarcasm or anything that might be misunderstood.

Highlight what you want: Be clear! Do you want a serious relationship or a casual fling? Honesty attracts matches on the same page.

Share a fun fact: Maybe you’ve traveled to 20 countries or once met a celeb in an unexpected place. These tidbits are great icebreakers.

Stay positive : Focus on what you enjoy. Avoid negative statements or listing things you don’t like — it’s far too Debbie Downer.

Your “About Me” section is your dating profile’s elevator pitch. It’s gotta be snappy, sassy, and totally you. Here’s how to make that blurb irresistible:

Create a vivid snapshot: Forget boring lists — paint a picture. Describe a typical weekend or a fave memory.

Share your aspirations: Talk about your dreams and goals. Are you working towards a marathon, planning to write a book, or desperately trying to master a new language?

Show your values : Highlight what’s important to you. Whether it’s pets, family, or personal growth, let folks see your core values to attract those on the same wavelength.

Add a touch of mystery: Don’t spill all your secrets. Hint at a story or drop a tantalizing detail to leave them wanting more. It’s all about the tease.

Highlight recent adventures: Talk about your latest escapades. Recently back from backpacking in Bali? Tried skydiving? Anecdotes make great convo starters.

Discuss cultural interests: What’s on your bookshelf or playlist? Share your faves as a bonding point. Plus, it’s a sneaky way to show off your personality.

End with a question : A question like, “What’s the best book you’ve read this year?” or “Any travel tips for my next adventure?” encourages peeps to reach out.

Here are a couple of examples to get the ball rolling:

“Bookworm by day, Netflix binge-watcher by night. I love curling up with a good novel or debating plot twists over a glass of wine. If you know a killer thriller or have a strong opinion on ‘Stranger Things,’ let’s chat!”

“Aspiring chef with a passion for trying new recipes and hunting down the best food trucks. If you’re into food adventures and don’t mind a little kitchen chaos, swipe right and let’s cook up some fun together!”

Avoid these common pitfalls to make sure your profile sends the right message:

Negativity: Don’t focus on what you don’t want. Life satisfaction can be attractive !

Desperation: You’re a catch, so avoid looking desperate !

Clichés: Avoid overused phrases like “I live life to the fullest.”

Personal deets: Keep things like your exact job location and home address private.

Demands : Making too many specifications about who you’re looking for can seem rigid.

The photos you choose for your dating profile can make or break your online dating success. Here’s what attracts the most positive attention:

High-quality photos : Clear, well-lit images are crucial.

A variety of shots : Include a mix of headshots, full-body images, and activities.

Authenticity : Photos should reflect your true self and lifestyle.

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Ultraprocessed Foods As Addictive As Tobacco, Researchers Say

Healthline

New research shows that ultraprocessed foods may be as addictive as cigarettes. Image Credit: Alexander Spatari/Getty Images

A recent review found that ultraprocessed foods (UPFs) may be as addictive as tobacco products.

The researchers found that UPFs are designed to heighten reward and accelerate the delivery of reinforcing ingredients.

The “addictive” quality of these foods means they drive compulsive consumption and disrupt appetite regulation.

The review authors suggest that ultraprocessed foods should be subject to regulations similar to those for tobacco products.

Currently, there is no single universal definition of ultra-processed foods. Some people define them as foods that contain ingredients you would not find in your kitchen cupboards, such as emulsifiers and additives.

Many professionals use the NOVA classification to define ultraprocessed foods. This talks about foods that contain “formulations of ingredients, mostly of exclusive industrial use, typically created by a series of industrial techniques and processes.”

A recent review published in The Millbank Quarterly suggested that ultraprocessed foods may be as addictive as tobacco products.

“I agree with this study, as ultra-processed foods are specifically engineered to be highly appealing to most individuals,” Mir Ali, MD, bariatric surgeon and medical director of MemorialCare Surgical Weight Loss Center at Orange Coast Medical Center in Fountain Valley, CA, who was not involved in the research, told Healthline.

Research from 2023 estimates that over 73% of the foods in the United States are ultraprocessed.

“Cigarettes and UPFs [ultraprocessed foods] are not simply natural products but highly engineered delivery systems designed specifically to maximize biological and psychological reinforcement and habitual overuse,” noted the new study’s research team from Harvard, Duke, and the University of Michigan.

Are ultraprocessed foods addictive?

The review synthesized findings from addiction, public health history, and nutrition in order to identify sensory and structural features that increase the reinforcing potential of both cigarettes and ultraprocessed foods.

“Not everyone is ‘addicted’ to these foods, but for a meaningful minority, they trigger classic addiction-like patterns: strong cravings, loss of control, and continued use despite negative health effects,” said Michelle Routhenstein, preventive cardiology dietitian at EntirelyNourished, who was not involved in the study.

“These foods are deliberately engineered with refined carbs, added fats, salt, and flavor enhancers to maximize reward and repeat consumption, and the industry uses aggressive marketing tactics, especially targeting children,” she told Healthline.

The researchers focused on five key areas:

delivery speed

hedonic engineering, or designing foods to be irresistibly good

dose optimization

environmental ubiquity

deceptive reformulation, or “health washing”

They found that, like cigarettes, ultraprocessed foods are fine-tuned to deliver the right dose of sugar.

“Refined carbohydrates stimulate dopamine release via the vagus nerve, whereas fats do so through intestinal lipid sensing,” the researchers wrote.

“... UPFs with high levels of refined carbohydrates and added fats are some of the most potently rewarding substances in the modern diet. Notably, this refined carbohydrate-fat combination is almost nonexistent in nature.”

The way these foods rapidly deliver “feel-good” chemicals to the brain makes them potentially addictive, similarly to cigarettes.

The review noted that while cigarettes are engineered to deliver nicotine quickly, ultraprocessed foods are engineered to rapidly digest and absorb, as they typically have little to no fiber. This makes it easier for the body to process fat and sugar more quickly.

The researchers also explain that ultraprocessed foods give intentional flavor bursts that fade quickly and textures that melt in your mouth. This all helps to deliver more dopamine and encourages you to eat more.

‘Health washing’ in cigarettes, ultraprocessed foods

According to the review, both the tobacco and food industries have long used a strategy called “health washing.”

This is where products are marketed and reformulated to create an illusion of reduced harm while preserving the core of their addictive properties.

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standard-article(title: [The Best of 2025], author: [Lindsay Landis], source-name: [Love and Olive Oil], images: (), paragraphs: ([See ya, 2025. Don't let the screen door hit you on the way out! (Seriously though, what a year!)], [The biggest milestone of 2025—for us at least—was the launch of Fresh Baked Puns , finally putting some of my favorite food puns on tshirts, tea towels, stickers and more (psst! If you missed it, I just added Live Laugh Loaf totes and aprons to the shop! The aprons in particular are extremely limited in quantity, so if you want one, I wouldn't wait too long.)], [I'm currently in the sketching phase for the next two designs coming very soon... let's just say you better be on the mailing list if you want first dibs.], [Anyway. I've done a best of post every year for the past 16 years (!) and I'm not about to stop such a long-standing tradition despite the fact my brain is half mush—I mean, cookie dough—at this point in the year.], [As for what's in store for 2026... I already have a few recipes in the works that I know you're going to love. I just need to hunker down and photograph the darn things. Then edit the photos. Then write the recipe and the post. Then promote it... (seriously though, every post takes hours and hours to pull together these days, no wonder I'm not publishing as frequently as I used to).], [And while I've never been the resolutions type, I've got a few big goals in addition to trying to develop some better habits this coming year, including:], [Draw everyday.], [Bake more homemade bread.], [Make more homemade pasta.], [Learn to shoot with artificial light (I'm so sick of fighting the sun).], [Release at least 4 new pun designs (one every quarter).], [Figure out ways to better market myself and Fresh Baked Puns (marketing expert I am not).], [Get more recipes converted, including metric measurements (I've made it back to 2019... still 12 more years—and 818 more recipes—to go!)], [Eat less sugar ehheh who are we kidding. That sweet tooth isn't going anywhere!], [What about you? What are you most excited for in the new year?], [(Click through to see which recipe came out on top!)], [(more...)],), insert-map: (:), word-count: 362, edited-for-length: false, debug-mode: false,)

standard-article(title: [The Graveyard That Made Me Kiss a Frog], author: [Louise Story], source-name: [Atlas Obscura], images: (), paragraphs: ([I kissed a frog in Sparta, Wisconsin. Voluntarily. Enthusiastically.], [The frog in question is one of hundreds of giant fiberglass molds scattered across a football-field-sized lot behind a nondescript sheet-metal building off County Highway Q. This is the home of FAST — Fiberglass Animals, Shapes, and Trademarks — a company that has been building giant roadside statues, mascots, and water park attractions since the early 1970s. It was incorporated under its current name in 1983 by a man named Jerome Vettrus.], [FAST has worn the mantle of American titan-builder for over 50 years. Among their greatest hits: a 200-foot-long sea monster at House on the Rock in Spring Green, Wisconsin, and a 145-foot-long muskie in Hayward.], [After each job, they keep the mold. All of them. For decades.], [And that's how a quiet field in rural Wisconsin became one of the most unexpectedly wonderful places I've ever wandered.], [There are giant skulls and colossal dogs, oversized Santa Clauses and titanic mice. The fiberglass has weathered over time, giving the molds an almost ancient, stone-like quality, as though the yard is the remnant of some surreal lost civilization. Walking through it is eerie and beautiful at the same time. Some molds are rotted out, covered in weeds or standing water. Others are in relatively pristine condition and could practically be reused tomorrow.], [That's actually the point. The molds are kept for future reuse, since they'd be expensive to recreate. So this isn't just a graveyard. It's also a library. A catalog. An archive of American roadside whimsy sitting in the tall Wisconsin grass.], [I found the frog slide mold and, yes, I kissed it (still looking for my prince). FAST has been making frog slides for over 35 years. You've almost certainly seen one at a water park somewhere without knowing it.], [I slid down a few of the slides too, because how could you not. Then I stood there imagining all the places these forms have traveled, whether they were painted bright yellow or fire-engine red, whether they were installed at some mini golf course in Arizona or a splash pad in Ohio. I imagined the faces of the delighted kids who have no idea their beloved frog came from a field in Wisconsin.], [The current owner took over around 2020, and if you visit during business hours, he might just show you around. The whole thing is free. Open 24 hours. No facilities. Pure wonder.], [It got me thinking about waste and beauty. There's something philosophically satisfying about a place where industrial byproduct becomes accidental art installation. FAST's mold graveyard isn't the only example. Ghanaian artist El Anatsui famously creates vast, shimmering tapestries from discarded bottle caps. Artists on Mount Everest have turned abandoned oxygen cylinders and helicopter wreckage into sculpture. The stuff we discard has a strange afterlife when someone thinks to look at it differently.], [That's the gift of Atlas Obscura exploring: we keep pointing you toward the places where someone already did the looking for you.],), insert-map: (:), word-count: 500, edited-for-length: false, debug-mode: false,)

And while I've never been the resolutions type, I've got a few big goals in addition to trying to develop some better habits this coming year, including: Draw everyday.

Lindsay Landis

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Roman Altars and Tombstones in Talavera de la Reina, Spain

Atlas Obscura

In the heart of Castilla-La Mancha, Talavera de la Reina makes a compelling case for a detour. Known for its brilliant blue-and-white ceramics and its easygoing riverfront along the Tagus, the city blends small-town charm with layers of deep history. Wander its tiled plazas, linger over local cuisine, and you'll quickly sense that Talavera rewards travelers who look closely-especially those willing to peer at its walls.

A short stroll brings you to the *Basílica de Nuestra Señora del Prado* ("Basilica of Our Lady of the Meadow"), an ancestral sanctuary whose eastern façade doubles as an open-air archaeological gallery. Embedded directly into the stone are medieval slabs and far older Roman altars and funerary inscriptions, each labeled with its original findspot somewhere in town. It's the kind of detail you might miss-unless you know to look up.

One small plaque mentions the "mares infernales," likely a slip for "manes," the benevolent spirits of the Roman dead. Another altar, though classified as Roman, resembles an Iberian Late Bronze Age stela, hinting at even deeper roots beneath the city's streets.

Over centuries, the basilica has absorbed donations of art and artifacts: the tombstone of Liborio, heraldic shields, Latin epigraphy, and a 15th-century Virgin and Child. In Talavera, history isn't confined to museums-it's mortared into the walls.

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BRIEFS

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The Sentinel

Vol. 1, No. 042 · 2026-03-30

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